

Weighting ADD

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There are three different types of weights (floats from 0 to 1):

- Tag Grouping Weight
 - o Social (Charisma, Rhetoric, Etc...): **0.75**
- Tag Weight
 - o Charisma (Body language, amiability, etc...): **0.95**
- Task Weight
 - o Need to learn body language: **0.66**

The higher the value of the float, the higher on the descending sorted queue it will be and considered for draw

User To Do List:

- Task1: 0.97
- Task2: 0.95
- Etc...

This means that in order to display accurate order, tasks need to have their weights updated on any task completion. Due to the dynamic tagging system, closing a task in a separate category can have an effect on a task in another.

Task weight is generated from tag weights, which are the only input necessary for proper weighting. Task weight is derived from taking the tests and is automatically updated on task completion.

Task Weight Scoring:

How to empathize:
↳ Connection: Kindness: 0.75, Love: 0.4 | $\bar{c}_{mean} = 0.575$
↳ Social: social: 0.9, Relationship: 0.95, Social awareness: 0.77 | $\bar{s}_{mean} = 0.873$
↳ ...

For each task, separate the tags into their respective tag groups. After this, mean the weight of those tags.

$$c_{tasks} = 2, s_{tasks} = 3, TotalTasks = c_{tasks} + s_{tasks} = 5$$
$$\bar{c}_{mean} * \frac{n_{tasks}}{totaltasks} = 0.575 * \frac{2}{5} = 0.23 ; \bar{s}_{mean} * \frac{3}{5} = 0.5238$$

Take the mean and multiple by the number of tags in that category, and divide by the total number of tags represented by the task.

$$c_1 \dots c_m = 0.23 + 0.5238 \dots = \boxed{0.7538} \rightarrow \text{How to empathize}$$

Continue doing this process for the other categories and sum the total. That is the composite weight for the task.

This process will be done every time a task is complete, because it is important in updating the tag weights and the tag group weights.

Updating Tag Weights:

$$Empathy = 0.75 \quad u * \left(1 - \frac{c}{N}\right)^\beta \rightarrow \begin{cases} c = \text{number of tasks complete for tag} \\ N = \text{number of total tasks} \\ \beta = \text{bias factor (1/4)} \\ u = \text{score} \end{cases}$$
$$\rightarrow 0.75 * \left(1 - \frac{3}{11}\right)^{1/4} = \boxed{0.6926} \rightarrow \text{updated score for empathy with 3 tasks done}$$

Take the initial score derived from the test and multiple by a factor reduced by the number of tasks completed over the total with a heavy bias factor. This ensures important scores are not decremented too much and still are a priority but allow for working on other things that have more work needed to be done.

Using this score we can again compute the Tag weight equation with a Tag Group to derive its score as seen below:

How To Be a good father

↳ Empathy: 0.75 — 15 tasks, 4 finished = 11 remaining
↳ financially set: 0.55 — 10 tasks, 2 finished = 8 remaining
↳ Patient: 0.65 — 4 tasks, none finished = 4 remaining

17 000



$$\text{total} = 11 + 8 + 4 = 22$$

tasks remaining

$$\text{Good Father Score} = \left(\text{Tag score} * \left(\frac{\text{tag remaining}}{\text{total remaining}} \right) \dots \right)$$

$$\hookrightarrow \left(0.75 * \left(\frac{11}{22} \right) \right) + \left(0.55 * \left(\frac{8}{22} \right) \right) + \left(0.65 * \left(\frac{4}{22} \right) \right)$$

$$\hookrightarrow 0.375 + 0.2 + 0.118$$

$$\hookrightarrow 0.493 - \text{good father score}$$

The tag weight coefficient scores are taken from the test using a sigmoid function to ensure float value between 0 < x < 1. The various tests have the following effects on their respective tags:

- Archetype: factor of 1 applied to entire relevant tag group
- Main goal: factor of 1 applied to every relevant tag
- Lifestyle: factor of 0.5 applied to every relevant tag
- Outlook: factor of 0.5 applied to every relevant tag
- Values: factor of 0.75 applied to every relevant tag
- Political background: factor of 0.25 applied to every relevant tag

Any tag associated with any of these categories will be then be processed by a sigmoid function to determine their float value.

Examples below:

Archetype results:

- Introvert tags +1
- Emotional tags +1
- Spontaneity tags +1
- Pragmatic tags +1

Main goal:

- Empathy +1
 - Emotional Intelligence +1
- 000

Lifestyle:

- Empathy +0.5
- Flexible +0.5

000 Political:
Modernism +0.25

All tags:

- Empathy_{total} = 1.5
- Emotional Intelligence_{total} = 1
- Flexible_{total} = 0.5
- Modernism_{total} = 0.25

$$\text{Sigmoid} = \frac{\text{Score}}{1 + \text{Score}} = \text{result}$$

$$\left\{ \begin{array}{l} \text{Empathy} = \frac{1.5}{1.5 + 1} = 0.6 \\ \text{EI} = \frac{1}{1 + 1} = 0.5 \\ \text{Flex} = \frac{0.5}{1 + 0.5} = 0.33 \\ \text{mod} = \frac{0.25}{0.25 + 1} = 0.2 \end{array} \right.$$